

EHC Meeting 9/15/26

Cooper Jackson



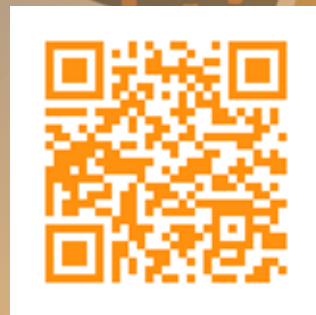
# LANL Summer School Internship | #ad

Cyber Incident Investigation School

10 week long program

- Each class taught by industry experts, along with CTF style puzzles. At the end you are provided with a simulated data set that you investigate and present to C-Suite

Network Archaeology -> Host Forensics -> Malware Analysis -> OT ->  
Simulated Incident Response



<https://cyberfire.energy.gov/school/>

# Upcoming Career Fair

LANL Information Session 9/16 in shelby 5pm.

LANL will be at the Engineering Career Fair this week (9/17)

Applications are open as of yesterday for the program.

- Admission is on a rolling basis (so apply asap if interested)
- Feel free to reach out about questions.



# CTF Puzzles

Throughout the summer, classes would be followed by CTF challenges that reinforced class content.

## 04\_dynamic

- 1-Lab – Basic Dynamic Analysis
- 2-HTTP Request-Response Code
- 3-File Written by Malware
- 4-Process Write
- 5-New Process Launched
- 6-Parent Process
- 7-Mutexes
- 8-Registry Key Persistence
- ☆

## 05\_assembly

- 1-Assembly With Godbolt-Intro
- 2-Assembly With Godbolt – Conditional Branches
- 3-Assembly With Godbolt – Control Variable Increment
- 4-Assembly – Understanding Documentation
- 5-Assembly – Registers
- 6-Assembly – Return Values
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## 06\_debugging

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## 07\_malware\_techniques

- 1-Malware-Techniques-1
- 2-Malware-Techniques-2
- 3-Malware-Techniques-3
- 5-Malware-Techniques-5
- 6-Malware-Techniques-6
- 7-Malware-Techniques-7
- 8-Malware-Techniques-8
- 9-Malware-Techniques-9
- 10-Malware-Techniques-10
- 11-Malware-Techniques-11
- 12-Malware-Techniques-12
- 13-Malware-Techniques-13
- 14-Malware-Techniques-14
- 15-Malware-Techniques-15
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## 10\_apheleia

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## 11\_crackmes

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## Safe\_Malwar

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- 2001
- 2002
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- 2004
- 2005
- 2006

## s2\_ai

- 101-AI-Lab-Intro
- 102-AI-Lab-crackme200
- 103-AI-Lab-crackme300
- 104-AI-Lab-crackme400
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## s2\_ttd

- 101-TTD
- 201-TTD
- 202-TTD
- 203-TTD
- 204-TTD
- 205-TTD
- 206-TTD
- 207-TTD

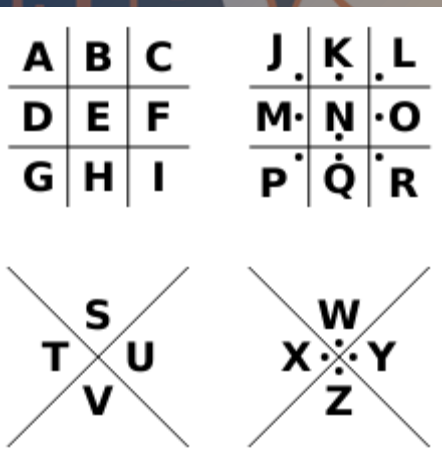
## s2\_unpacking

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# Network Archaeology

Teaches techniques to extract undocumented protocol communications

- Reverse engineering protocols straight from packet capture (PCAP)
- Writing custom decoders
- Wireshark, Zeek, Arkime, Scapy
- Directly used later to decrypt Cobalt Strike and Metasploit traffic



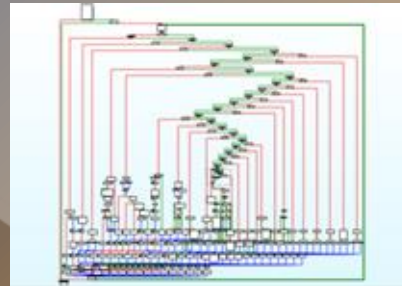
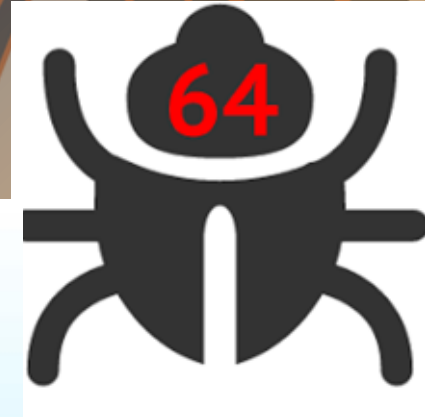
# Host Forensics

## Analysis of memory and drives

- Memprocfs, volatility, hayabusa, ftk imager, autopsy, etc.
- Hayabusa collapses thousands of Windows event logs into one readable timeline
- The Cobalt Strike beacons were fileless – memory only, never written to disk

# Malware Analysis

- 2 weeks long
- In depth look at Windows Malware.
- Ghidra, IDA, x64dbg,
- Utilizing AI to Reverse Engineer (Ghidra MCP Server)
- static/dynamic analysis, Packed, Anti-debugging, windows api calls, obfuscated calls.



# Operational Technology (OT)

The systems that monitor and control physical devices

- Hands-on with the Controller Area Network (CAN) bus protocol on real hardware
- Industrial Control Systems (ICS): Programmable Logic Controllers (PLCs), Supervisory Control and Data Acquisition (SCADA)

# Context on the Project Dataset:

Synthetic Dataset with a hypothetical Incident Response Scenario.

In this case, Election Fraud of an government IT/OT voting system network

Provided data:

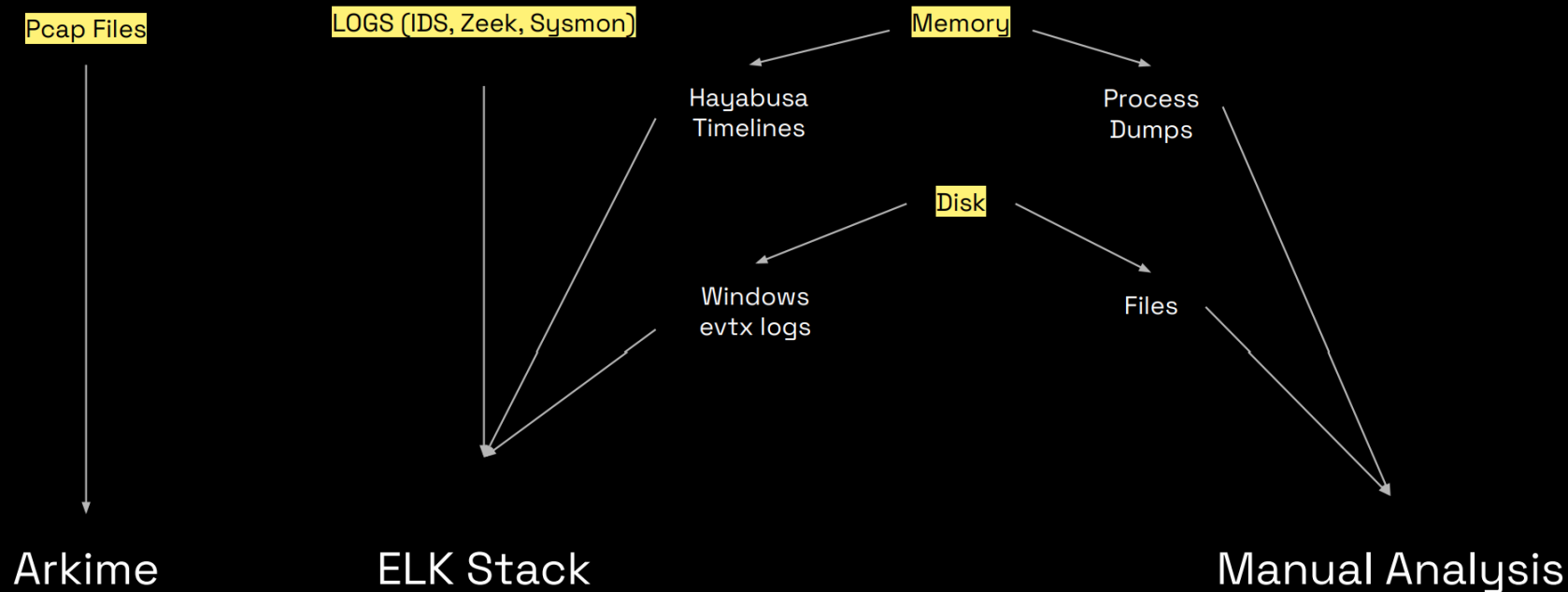
Disk Images of systems (11 total; 9 windows OS, 1 Linux, 1 Mac)

Memory Images (30+, multiple copies of same systems at different times throughout the incident)

Pcap files of traffic. (80+ GB of network traffic.)



# Tooling and Data



# Some Vocab

- Command and Control (C2) – the attacker's remote server; the malware calls home to it for orders
- Indicator of Compromise (IOC) – a file hash, IP, domain, or filename that proves the attacker was there
- Lateral movement – hopping host to host with stolen credentials once inside
- Persistence – surviving a reboot, via scheduled tasks, services, or startup scripts
- Exfiltration – stealing data out of the network, often hidden inside normal-looking web traffic
- Dual-homed host – one machine on two networks at once; our attacker's bridge into OT
- Process injection – running malicious code inside a legitimate Windows process so it looks normal

# Onto the C-Suite Presentation

